Applications of AI in Africa (*AI-Based System for Malaria Surveillance in Africa Using Mosquito Wingbeat Vibrations*): A Proposal

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I. Introduction (*or* Project Overview)

Malaria is a life-threatening infectious disease caused by Plasmodium parasites and transmitted to humans through the bite of infected female Anopheles mosquitoes. The disease is most prevalent in tropical and subtropical regions characterized by high humidity, warm temperatures, abundant rainfall, and agricultural activities that create stagnant water, providing suitable breeding habitats for mosquitoes [1]. Common symptoms include fever, chills, and headache, while severe cases may result in seizures, fatigue, respiratory distress, confusion, and, if left untreated, death [2].

According to the World Malaria Report 2025 of the World Health Organization (WHO), Africa accounted for approximately 95% of global malaria cases and deaths in 2024, with nearly one child under the age of five die of malaria every minute [2] [3].

Despite significant progress in malaria control through interventions such as vaccination, the use of insecticide-treated mosquito nets, indoor residual spraying, and other vector-control strategies, malaria continues to pose a major public health challenge across many African countries. Consequently, there is a growing need for intelligent surveillance systems capable of identifying areas at elevated risk of malaria transmission, thereby enabling public health authorities to implement targeted interventions before outbreaks escalate.

This project proposes the development of a machine learning model that analyzes the wingbeat vibration signatures of mosquitoes to identify Anopheles species and estimate their abundance within a given region. By using mosquito acoustic data as a proxy for vector surveillance, the proposed system aims to predict the likelihood of malaria transmission in tropical and subtropical regions, providing a cost-effective and scalable tool to support early warning systems and informed public health decision-making.

II. Problem Statement

Malaria remains a major public health challenge in Africa, accounting for a substantial burden of illness and mortality, particularly in sub-Saharan Africa.

Although global malaria case incidence declined by 25.6% between 2000 and 2015 (from 79.4 to 59.0 cases per 1,000 population at risk), progress has stalled and partially reversed. Between 2015 and 2024, malaria case incidence increased by 8.5%, rising from 59.0 to 64.0 cases per 1,000 population at risk [2].

This resurgence highlights the need for effective surveillance methods capable of monitoring mosquito vector abundance to support the early identification of areas at elevated risk of malaria transmission.

III. Proposed Solution

This project proposes the development of an AI-powered mosquito surveillance system that uses machine learning to classify mosquito species from their wingbeat acoustic signatures. The model will be trained and validated using a combinations of a dataset from Kaggle [4] and the HumBugDB acoustic dataset [5] to distinguish malaria vector species, particularly female Anopheles mosquitoes, from non-vector mosquito species. The model will then be quantized to be deployed on any microcontroller preferably a Raspberry Pi.

Using Kenya and Tanzania as a case study, the trained model will analyze mosquito recordings collected from a given location to estimate the relative abundance of major malaria vector species, including Anopheles gambiae, Anopheles funestus, Anopheles stephensi, Anopheles arabiensis, and Anopheles pharoensis.

The proposed solution will take the form of a mobile application integrated with the trained machine learning model as a proof of concept, although the final quantized model will be deployed on sensors

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IV. Aim and Objectives

A. Main Objective

Develop an AI-based acoustic system to identify malaria-vector mosquitoes to support malaria surveillance.

B. Sub-objectives

1) Train and quantize a machine learning model to classify female Anopheles mosquitoes from other mosquito species.
2) Develop a mobile application interface that predicts mosquito species and visualizes malaria surveillance information.

V. Methodology

This project follows a standard machine learning process:

A. Data Preprocessing and Feature Extraction

Collect dataset containing audio recordings of mosquitoes wing beats and other features that play a key role in determining mosquito species. Clean the dataset before using it; remove unnecessary data to ensure that the model is trained on well organized data.

Turn the wingbeat recordings into a format that the model can learn from.

B. Model Development and Quantization

Train the model on the data set to classify Anopheles mosquito species, among others. We may use CNN, Random Forest, or SVM to ensure that we better classify mosquito species. The best model will be selected on the basis of accuracy. The selected model will then be quantized to evaluate its suitability for deployment on resource-constrained devices and to assess the impact of quantization on model accuracy and inference performance.

C. Risk Mappping

Pick the best performing model to map the model results with the region whose recordings belong. Determine the risk rate of malaria spread for each participating region.

D. Mobile Application

Develop an interface for proof of concept where users can upload mosquito sounds, view prediction results for the classification of mosquito species, and visualize mosquito surveillance information (risk rate of each participating region, and more).

E. List of Assumptions

1) The presence of female Anopheles mosquitoes is used as a proxy indicator of the potential risk of malaria transmission.

2) The audio data in both the HumBugDB acoustic mosquito dataset and the kaggle dataset is sufficient and reliable to train an ML model accurately.
3) The developed ML model can be used to identify malaria vector species as a proxy to estimate transmission risk in Africa.

F. List of Resources

Dataset: HumBugDB acoustic mosquito dataset from Zenodo and Kaggle [5] [4]

Programming Language: Python

Machine Learning Libraries: Scikit-learn, TensorFlow

Other Tools: VSCode, Google Colab, GitHub

VI. WORK PLAN

Milestone	**Responsible**	**Timeframe**
Proposal Writing	Hana Andarge, Delice Ishimwe, Janet Uwamahoro	Aug 3rd-7th
ML Model Development and Quantization	Elvis Guy Bakunzi, Janet Uwamahoro, Trevor Tebaweswa	Aug 8th-14th
Frontend Development	Delice Ishimwe, Emmanuel Chigere, Hana Andarge	Aug 8th-14th
Report Writing	Janet Uwamahoro, Delice Ishimwe, Elvis Guy	Aug 14th-17th
Presentation and Slides	Emmanuel Chigere, Hana Andarge, Trevor Tebaweswa	Aug 14th-17th

TABLE I: Team Roles and Responsibilities

VII. AI USE DISCLOSURE

AI was used to polish the language used in some parts of the report for sentence structure and word choice.

REFERENCES

[1] CDC, “Where malaria occurs,” 2024. [Online]. Available: https://www.cdc.gov/malaria/data-research/index.html

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[5] Kiskin, Sinka, Cobb, Rafique, Wang, Zilli, and Gutteridge, “Humbugdb: A large-scale acoustic mosquito dataset,” *Zenodo*, 2021.